

Lab 5

Exercise 1

A car rental company has hired you to design a relational schema to hold their booking's data. Customers provide their details as well as the type of car they want to rent and any extra services they would like to include in the booking (for example GPS or child seat). A sample of their data is shown on Table 1. Normalise the system to the third normal form.

BookingID	Car Type	base rent	custID	Name	DOB	Age	Cust Address	Date of booking	Service name	service charge
1	Sedan	50	101	John	1990-05-15	33	123 Main St, City	2023-10-17	GPS, Child Seat	10,15
2	SUV	80	102	Sarah	1985-07-21	38	456 Elm St, Town	2023-10-16	Extra Driver	15
3	Hatchback	40	103	Michael	1995-03-10	28	789 Oak St, Village	2023-10-15	Wash & Vacuum	20
4	Convertible	70	104	Emily	1988-11-30	34	101 Pine Rd, County	2023-10-14	GPS, Wash & Vacuum	10,20
5	Truck	90	105	Daniel	1980-09-05	43	222 Cedar Ln, City	2023-10-13	GPS	10
6	Sedan	50	106	Jessica	1992-04-20	31	333 Birch Ave, Town	2023-10-12	Child Seat	15
7	Hatchback	40	107	Brian	1987-08-12	36	444 Oak Rd, City	2023-10-11	Wash & Vacuum	20
8	SUV	80	108	Amanda	1993-01-25	30	555 Elm St, Village	2023-10-10	Child Seat	15
9	Convertible	70	109	Robert	1997-06-03	26	666 Maple Ln, County	2023-10-09	GPS, Extra Driver	10,15
10	Sedan	50	110	Olivia	1996-12-18	26	777 Pine St, Town	2023-10-08	Wash & Vacuum	20

Table 1 Car Rental Data

Step 1 Convert the table into First Normal Form (1NF):

No Repeating Elements or Groups of Elements.

Clearly identify the primary key(s).

Step 2 Second Normal Form (2NF): No Partial Dependencies on a Concatenated Key

For Second Normal Form (**2NF**) each table must be tested for *partial dependencies* on a *concatenated key*. This means that for a table that has a concatenated primary key, each column in the table that is not part of the primary key must depend upon the entire concatenated key for its existence. If any column only depends upon one part of the concatenated key, then the entire table has failed 2NF and you must create another table to rectify the failure.

Step 3 Third Normal Form: No Dependencies on Non-Key Attributes

Each column that relies on 'non-key attributes' now needs to be considered and all transitive dependencies removed.

Draw the **final 3-NF Entity Relationship Diagram**.

Exercise 2

Consider a company keeping track of projects, employees and departments. Table 2 contains the information stored by the company. Convert this table into a db in 3NF.

General Requirements

- Each project has a unique name.
- Employee and manager names are not unique
- Many employees can work in each project and each employee can be assigned to more than one project. Hours refers to the number of hours and employee is assigned to a particular project.

- Budget and start_date refer to the total budget for the project and project start date.
- Salary refers to the employee's salary
- Empl_manager refers to the name of the employee's manager, which may be different to the project's manager.
- Empl_department refers to the employee's department. Department names are unique and the employee's manager is the employee's department manager.
- Rating refers to the employee's performance in a specific project.

projName	projManager	Empl_id	hours	Employee name	budget	Start_date	salary	Empl_manager	Empl_department	rating
Gamma	Smith	101	25	Jones	200000	01/01/2021	2000	O'Reilly	10	9
Gamma	Smith	121	32	Lee	200000	01/01/2021	2500	Kennedy	12	
Gamma	Smith	104	40	Kelly	200000	01/01/2021	1000	O'Reilly	10	6
Delta	Murphy	101	25	Jones	500000	02/10/2021	2000	O'Reilly	10	6
Gamma	Smith	133	32	Lee	200000	01/01/2021	2500	Kennedy	12	
Delta	Murphy	111	15	Bowles	500000	02/10/2021	1500	Kennedy	12	7

Table 2 Company Table

WORK(proj_Name, proj_manager, empl_id, hours, empl_name, budget, startDate, salary, empl_manager, empl_dept, rating)